

Groups and Presentations of Finite Groups

MATH200

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Unary Operation

A **unary operation** u on a set A is a function $u : A \rightarrow A$.

Binary Operation

A **binary operation** $*$ on a set G is a function $*$: $G \times G \rightarrow G$. For any $a, b \in G$ we shall write $a * b$ for $*(a, b)$.

Algebraic Structure

An **algebraic structure** is a set G together with a finite number of binary and unary operations defined on G . If $*_1, \dots, *_r$ and $u_1, \dots, u_s$ are the operations that define algebraic structure on G , then we denote the structure by $(A, *_1, \dots, *_r, u_1, \dots, u_s)$.

Associativity

A binary operation $*$ on a set G is **associative** if $\forall a, b, c \in G$ we have:

$$a * (b * c) = (a * b) * c$$

Commutativity

If $*$ is a binary operation on a set G we say elements a and b of G commute if $a * b = b * a$. We say $*$ (or G) is **commutative** if $\forall a, b \in G$,

$$a * b = b * a.$$

Identity Element

Let $*$ be a binary operation on a set G . There exists an element $e \in G$, called an **identity** of G , such that $\forall a \in G$ we have:

$$e * a = a * e = a$$

Inverse Element

Let $*$ be a binary operation on a set G . $\forall a \in G$ there exists an element $a^{-1} \in G$, called **inverse** of a , such that

$$a^{-1} * a = a * a^{-1} = e$$

where e is the identity element.

Homomorphism

A homomorphism is a function between two structures of the same type that preserves operations of structures. Let A_1, A_2 be two sets and $*_1$ and $*_2$ be two binary operations. $(A_1, *_1)$ and $(A_2, *_2)$ be same type of algebraic structures. A function $\varphi : A_1 \rightarrow A_2$ is a **homomorphism** if $\forall a, b \in A_1$

$$\varphi(a *_1 b) = \varphi(a) *_2 \varphi(b)$$

Isomorphism

If a function is both a homomorphism and a bijection then we can call it a **isomorphism**.

Structures That Are Simpler (!) Than Groups

Magmatic Structures

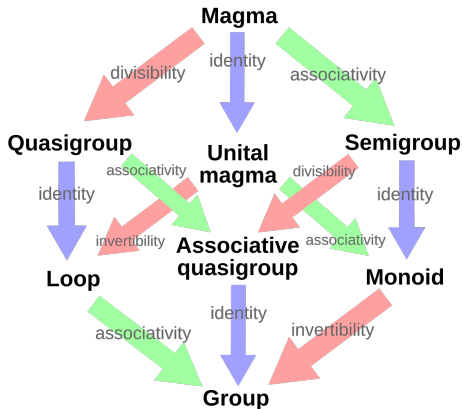


Figure: Diagram of Magmatic Structures

Introduction to Groups

Definition (Group)

Let $*$ be a binary operation and G be a non-empty set, we have three laws for $(G, *)$ to be a **group**:

G1: Binary operation $*$ is associative.

G2: $\exists e \in G$ such that $\forall a \in G, a * e = e * a = a$, i.e. e is the identity element.

G3: $\forall a \in G$ there exists $a^{-1} \in G$ such that $a * a^{-1} = a^{-1} * a = e$ where e is the identity element.

Definition (Abelian (Commutative) Group)

The group $(G, *)$ is called "**Abelian**" (or commutative) if $a * b = b * a$ for all $a, b \in G$.

Definition (Subgroup)

A **subgroup** of a group $(G, *)$ is a subset $H \subseteq G$ that forms a group under same operation as G .

Theorem

Let G be a group and let $a \in G$. Then

$$H = \{a^n \mid n \in \mathbb{Z}\}$$

is a subgroup of G and is the smallest subgroup of G that contains a , i.e. every subgroup containing a contains H .

Definition (Cyclic Subgroups)

Let G be a group and let $a \in G$. Then the subgroup $H = \{a^n \mid n \in \mathbb{Z}\}$ of G is called the **cyclic subgroup** of G generated by a and denoted by $\langle a \rangle$.

Definition (Cyclic Group)

An element a of a group G generates G and is a generator for G if $\langle a \rangle = G$. A group G is **cyclic** if there is some element $a \in G$ that generates G .

Theorem

Every cyclic group is abelian.
A subgroup of a cyclic group is cyclic.

Elementary Examples of Groups

- $(\mathbb{Z}, +)$
- $(\mathbb{R}, +)$
- $(\mathbb{Z}_n, +)$
- $(\mathbb{Q}/\{0\}, \cdot)$

Historical Motivation of Groups

- What is the purpose of groups?
- Does "Group Theory" appeared out of vacuum?
- Did it started as Group Theory?

History of Groups: Lagrange



Figure: Joseph-Louis Lagrange 1736-1813 (Retrieved from *Wikimedia*)

An informal historical introduction to Group Theory.

- Start of an idea
- Precursor for better understanding of polynomial equations algebra
- First idea of group of permutations for equation theory.

History of Groups: Galois

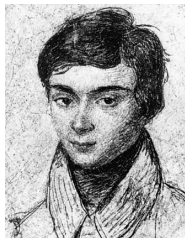


Figure: Evariste Galois 1811-1832 (Retrieved from *Wikimedia*)

- First use of "*groupe*"
- Galois Groups / Permutation Groups
- Group Theory + Field Theory = Galois Theory
- Normal Subgroups
- Solvability

Later on History: Cauchy, Jordan, Cayley, Klein...

Augustin Cauchy

- More words over "permutation groups"
- Works of Galois

Camille Jordan

- Idea of isomorphism
- Groups in general

Arthur Cayley

- Cayley Diagram
- Finite Groups
- Cayley's Theorem

Felix Klein

- First ideas of groups in other areas of Mathematics
- Geometry of Groups $\rightarrow$ Erlangen Program
- Symmetry Groups

Examples of Finite Groups

$\mathbb{Z}_n$ ($\mathbb{Z}/n\mathbb{Z}$)

This is the group of equivalence classes over $\mathbb{Z}^+$. For example $\mathbb{Z}_5$ is the equivalence classes (mod 5):

$$\mathbb{Z}_5 = \{\bar{0}, \bar{1}, \bar{2}, \bar{3}, \bar{4}\}$$

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And $(\mathbb{Z}_5, +)$ is a finite, cyclic group.

Examples of Finite Groups

Easier understanding for Cyclic Groups and $\mathbb{Z}_n$

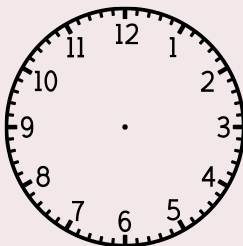


Figure: A clock without hands (Retrieved from *pngtree.com*)

Examples of Finite Groups

Permutation and Permutation Multiplication

We define permutation as a rearrangement over the elements of a set. But in its nature a permutation is a bijective function which maps set elements to its elements in the rearranged form. By the structure of it we can already see that a function composition is a binary operation. For the case of permutations, WLOG function composition is called permutation multiplication since it behaves like a non-commutative multiplication.

Theorem

Let A be a non-empty set, and let S_A be the collection of all permutations of A . Then S_A is a group under permutation multiplication. (Fraleigh, 2018, pp. 77-78)

Examples of Finite Groups

S_3

$$\begin{aligned}\rho_0 &= \begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix}, & \mu_1 &= \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix}, \\ \rho_1 &= \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix}, & \mu_2 &= \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix}, \\ \rho_2 &= \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix}, & \mu_3 &= \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix}.\end{aligned}$$

Figure: Permutations over the set $\{1, 2, 3\}$

S_3 Group

$$S_3 = \{\rho_0, \rho_1, \rho_2, \mu_1, \mu_2, \mu_3\}$$

Cayley Table and Presentations of a Finitely Generated Group

Cayley Table

A **Cayley Table** for a group G with a binary operation $*$ is a square table where:

- Each row and column corresponds to an element of G
- The entry in row x and column y is the product of $x * y$
- The table satisfies the group axioms: closure, associativity, identity, and inverses.

Connection with Finitely Generated Groups

Since finitely generated groups have finite elements, we can see all properties of a finite group in a Cayley Table, i.e. its presentation.

S_3 , D_3 and Their Presentations

Definition (Dihedral Group)

Dihedral Group D_n is the group of symmetries of the regular n -gon.

Example (D_3)

D_3 is the group of symmetries of the equilateral triangle.

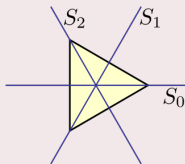


Figure: An equilateral triangle with symmetry lines shown
(Retrieved from *Wikimedia*)

S_3 , D_3 and Their Presentations

	ρ_0	ρ_1	ρ_2	μ_1	μ_2	μ_3
ρ_0	ρ_0	ρ_1	ρ_2	μ_1	μ_2	μ_3
ρ_1	ρ_1	ρ_2	ρ_0	μ_3	μ_1	μ_2
ρ_2	ρ_2	ρ_0	ρ_1	μ_2	μ_3	μ_1
μ_1	μ_1	μ_2	μ_3	ρ_0	ρ_1	ρ_2
μ_2	μ_2	μ_3	μ_1	ρ_2	ρ_0	ρ_1
μ_3	μ_3	μ_1	μ_2	ρ_1	ρ_2	ρ_0

Presentation of D_3

$$\rho_0 = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix}, \quad \mu_1 = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix},$$

$$\rho_1 = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix}, \quad \mu_2 = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix},$$

$$\rho_2 = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix}, \quad \mu_3 = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix}.$$

Elements of S_3 - D_3

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- Fraleigh, J. B. (2018). A first course in abstract algebra (8th ed.). Pearson
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